

Grant Trewartha

PhD · BSc · FHEA · AgilePM

Health-technology R&D leader bridging academia and industry. I move easily between big-picture strategy and hands-on data science - winning funding, leading teams, and turning rigorous research into things people actually use. A thinker and a doer.

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01 EXPERIENCE

2023 — Present Professor of Digital Health Interventions, Teesside University

- Leading concurrent multi-partner programmes spanning remote patient monitoring, AI clinical decision support and AI-informed triage, directing scope, delivery and stakeholder engagement across NHS, academic and commercial partners.
- Securing competitive research funding from the Academy of Medical Sciences, NIHR and EPSRC, setting strategy, creating workflows and coordinating the cross-disciplinary delivery.
- Co-Directing the Centre for Healthcare Innovation and REF co-lead for Allied Health, accountable for research quality, governance and external positioning.
- Industry consultancy for Visiba Ltd on “AI-triage in same-day urgent care”
- Developing talent by mentoring early career researchers and supervising doctoral candidates
- Building health technologies: remote monitoring solutions for gait and vital signs monitoring
- Associate Editor, Frontiers in Bioengineering & Biotechnology

2016 — 2023 Head of Research, NURVV Ltd (Consumer Electronics)

- Set and owned the R&D strategy for wearable technology products across sport and healthcare, repositioning the portfolio toward higher-value clinical applications.
- Built and line-managed the biomechanics function, embedding Agile delivery.
- Co-created sensor algorithms and app features, delivering six patents on “smart insole” technology.
- Directed external validation programmes with partner organisations and authored the propositions and pitches used to win investor and strategic partner backing.

2001 — 2016 Reader / Associate Professor in Biomechanics, University of Bath

- Founded and led the Sport Injury Prevention Research Group, growing it to 16 staff and delivering evidence that shaped international sports-governance policy and reduced injuries in rugby.
- Secured £1.5M in competitive research funding as principal investigator, with a body of work (85+ publications, 120+ presentations, H-index 34) establishing sustained external authority in the field.
- Consultancy activities included expert legal witness (knee injuries), coordinating “Catastrophic Injury Surveillance” registry for rugby, and “Smart Sports Impact Protection” report.
- Directed the BSc/MSci Sports Science programme and extensive project supervision: 13 doctoral, 20 masters and 80 undergraduate researchers. 15 years of curriculum design and lecturing experience.

02 EDUCATION & QUALIFICATIONS

Jan 2026

“AI in Healthcare” Executive Program

MIT Sloan School of Management

May 2016

Agile Project Management

APMG International

July 2001

PhD in Biomechanics

Loughborough University — “Marker-free Tracking of Human Motion”

July 1997

BSc Sports Science: First Class Honours

Loughborough University

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at home in strategy and in the data. A thinker and a doer.

03 CORE COMPETENCIES

● Originating & Winning Work

- Secured >£1.6M as principal investigator (Academy of Medical Sciences, NIHR, EPSRC, Rugby Football Union, World Rugby)
- Authored propositions that won investor and strategic partner backing at NURVV Ltd
- Confident stakeholder and investor liaison

● Leading Multidisciplinary Delivery

- Directed R&D programmes spanning NHS, academic and commercial partners
- Built and line-managed a research team of 16, embedding Agile delivery
- Cross-functional collaboration: product, engineering, firmware, app and marketing

● Healthcare Innovation & Applied AI

- Project managed deployment of multimodal AI triage in live NHS settings (Visiba Ltd)
- Six granted patents in wearable smart insole technology. UI/UX/QA testing experience
- ML/AI applied to clinical decision support and biomechanics

● Insight to Impact

- My research has shaped international sports-governance policy and reduced injuries
- Products taken from concept to market delivery in sport and healthcare
- Sustained publishing track record: 85+ articles, H-index = 33

04 PROFESSIONAL SKILLS

INDUSTRY KNOWLEDGE

R&D Leadership Agile PM

Wearable Technology

Remote Patient Monitoring

Management & Mentoring

Product Development

Media / Investor Pitches

Lecturing / Teaching

TECHNICAL & SCIENTIFIC

Python Matlab

R ML / AI

Multimodal AI Triage

Biomechanics Measurement

Signal Processing & Statistics

OpenSim: MSK Simulation

Claude Code

ADDITIONAL

Product QA & UX Testing

App Feature Logic & UI

Presentation Skills

Mobile App Development

Physical Computing

Blender / Maya

Sport Coach – Cricket & Football

05 ADVISORY & CURRENT WORK

Consultancy engagements and research programmes with impact

01

EPSRC NorthFutures

Frailty Virtual Wards

Remote monitoring (wearable biosensors) and AI clinical support (nearest neighbour analysis) to enable proactive intervention without hospital admission.

02

Industry – Visiba Ltd

AI Triage

Advising on multimodal AI triage for NHS care navigation - inbound data (vitals, symptoms, patient), shaping models, validation and clinical adoption.

03

Expert Witness

Injury Litigation

Expert witness in knee injury litigation, translating biomechanical evidence into decision-grade opinion for legal proceedings.

04

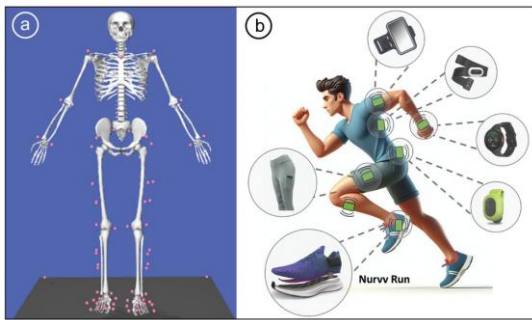
Sport Governance

Injury Surveillance & Prevention

Coordinated a national catastrophic injury surveillance registry for rugby. My rugby research ("Scrum Forces" and "Activate" programmes) was implemented internationally and has reduced catastrophic injuries.

<https://www.scopus.com/authid/detail.uri?authorId=22956376300>

06 HIGHLIGHT PUBLICATIONS



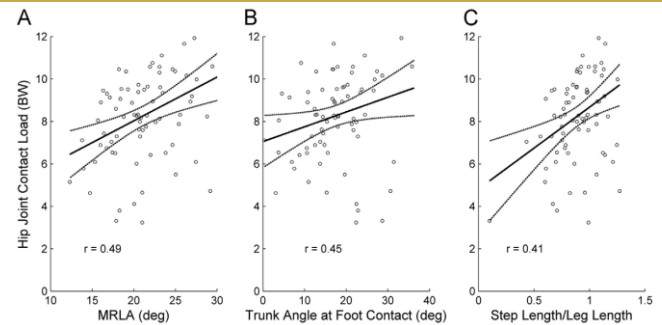
Sports Biomechanics

2025

Joint moments from wearable sensors & deep learning

DOI: 10.1080/14763141.2025.2526702

Neural network models estimated lower-limb joint moments in 50 runners; a wrist + foot sensor setup matched lab accuracy to within 10%.



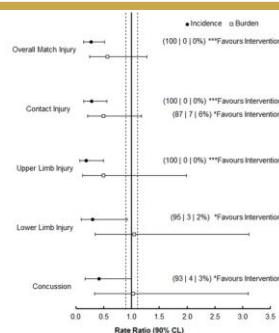
Journal of Biomechanics

2016

Hip contact loads during balance recovery in older adults

DOI: 10.1016/j.jbiomech.2016.05.033

Musculoskeletal modelling of 76 older adults showed peak hip loads up to 12.7 body-weights during stepping recovery - at times exceeding cadaver femur failure loads.



Br J Sports Medicine

2017

Movement-control programme reduces youth rugby injury

DOI:10.1136/bjsports-2016-097434

Cluster-RCT across 40 schools (3,188 players): teams using the warm-up programme 3 x week cut match injuries by 72% and concussions by 59% versus controls.



Scand J Med Sci Sports

2016

Pre-bind engagement improves scrum loading conditions

DOI: 10.1111/sms.12592

Live scrums across 27 teams: a pre-bind protocol cut peak engagement forces by up to 25% with no loss of push - evidence that shaped World Rugby law amendment.

The digital divide in respiratory care

BARRIERS TO ACCESS

- Socioeconomic & affordability
- Age & health limitations
- Low digital health literacy
- Geography (rural & coastal)
- One-size-fits-all design
- Trust & psychosocial factors



MULTI-SCALE SOLUTIONS

- National infrastructure & access
- Inclusive, co-designed tools
- Digital health literacy training
- Build trust with communities
- Cross-sector partnerships
- Measure benefit, not just access

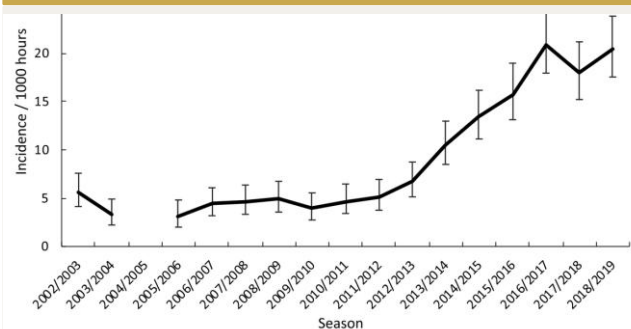
Chronic Respiratory Disease

2026

Digital exclusion in chronic respiratory disease

DOI: 10.1177/14799731261441307

Lead-author review mapping the barriers driving digital exclusion in CRD and the multi-scale, co-designed solutions so digital health narrows, not widens, inequity.



Brain Injury

2021

16-season trends in professional rugby concussion

DOI: 10.1080/02699052.2021.1972142

Prospective cohort of 3,006 players: diagnosed concussion incidence rose ~5-fold as recognition improved, while graduated return-to-play ended sub-6-day returns.